Project for the subject

INNOVATION & ENTREPRENEURSHI (UTA025)

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Group No.: **3C53_1**

Title: **Krishi Udaan**

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THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

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VENTURE LAB,

THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY, PATIALA

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CERTIFICATE

This is to certify that the project on, "**Krishi Udaan (3C53_1)**" being submitted by **Ms Jahanvi Varshney, Ms Yashvi, Ms Jiya Agrawal, Mr Anirudh Garg and Mr Arpit Jain** to the Venture Lab, Thapar Institute of Engineering and Technology, Patiala for the fulfilment of the course requirement of **INNOVATION & ENTREPRENEURSHIP (UTA025)** is a bonafide record of work carried out by us in conformity with the rules and regulations of the institute.

The results presented in this report have not been submitted, in part or full, to any other University or Institute for the award of any degree or diploma.

Dated: 25/04/2024

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Opportunity Analysis

1.1 Opportunity Identification

- In India, millions of farmers demand MSP (Minimum Support Price) to secure a fair income for their crops, but the government cannot offer MSP to all due to resource constraints.
- As a preventive safeguard, Krishi Udaan offers real-time crop monitoring and intelligent alerts to help farmers avoid losses from crop diseases, pests, and unpredictable weather.
- Farmers often lack formal agricultural education, leading to issues like improper fertilizer usage, delayed pest control, and inefficient irrigation.
- Locust attacks, droughts, floods, and other extreme conditions damage crops and incomes, especially in areas with limited access to support services.
- There's no widely adopted, unified system that helps farmers monitor crop health, plan preventive actions, and connect with advisors in real time.
- With rising smartphone penetration and government push for digital agriculture, there's a huge opportunity to empower farmers with accessible, AI-based tools for sustainable farming and risk mitigation.

1.2 Solution Proposed

- Early detection of diseases using AI-Powered Drones equipped with multispectral and thermal cameras to scan fields and detect early signs of diseases, nutrient deficiencies, pest infestations, and water stress.
- Farmers to receive real-time crop health reports, recommendations, and alerts in regional languages using mobile and web applications
- Intelligent Chatbot Support to provide instant, 24/7 guidance on fertilizer use, pest management, irrigation, and weather updates.
- Predictive Analytics & Smart Alerts based on collected data to help farmers take preventive action and reduce losses.
- Integration with Existing Farm Management Systems (like John Deere Operations Center) for enhanced data syncing and large-scale usage.
- Data-Driven Decision Support for farmers with actionable recommendations and historical crop performance tracking.
- Offline Accessibility and periodic sync to ensure usage in low-connectivity rural areas.

1.3 Uniqueness of the solution

- **Precision Monitoring using drones:** AI-integrated drones equipped with multispectral and thermal cameras enable early detection of diseases, pests, and crop stress, reducing yield loss and resource waste.
- **24/7 AI Chatbot Support:** An intelligent chatbot provides instant, accurate, and localized agricultural guidance on crop health, fertilizers, pest control, and irrigation, even for digitally unskilled users.
- **Real-Time Mobile and Web App:** A simple, multilingual interface allows farmers to access real-time insights, crop health analytics, and actionable recommendations from anywhere.
- **Predictive Analytics for Smart Decision-Making:** Data collected from drones and historical farm records is analyzed to give farmers proactive alerts and customized suggestions for better yield and efficiency.
- **Scalable, Offline-Ready, and Integration-Friendly:** Designed to function in lowconnectivity zones with offline sync and ready integration with popular farm management systems like the John Deere Operations Center.

1.4 What problem of the market segment is solved by your solution

- **Farmers often rely on instinct or delayed expert visits**: Krishi Udaan replaces this with AI driven, real-time crop health insights for timely decisions.
- **Pests, diseases, and nutrient issues silently destroy crops**: Our drone-powered detection system spots them early—before they become disasters.
- **Unpredictable rain, heat, or frost:** We send smart alerts that help farmers prepare, protect, and prevent losses due to sudden weather changes.
- **With our 24/7 chatbot in local languages, farmers don’t have to wait days for advice**: they get personalized guidance instantly, right from their field.

1.5 Identification of User and Customers

- **Individual Farmers:** Small, marginal, and large-scale farmers seeking reliable, data-driven crop monitoring and guidance.
- **Farmer Producer Organizations (FPOs):** Groups of farmers that can collectively use the platform to optimize yield and reduce losses across communities.
- **Agri-Tech Companies:** Businesses looking to integrate drone-based analytics and AI tools into their farming services.
- **Government Agriculture Departments:** Agencies promoting smart farming, sustainability, and rural development initiatives.
- **Agricultural Input Providers:** Companies selling seeds, fertilizers, and pesticides that can use Krishi Udaan’s data to offer targeted solutions.
- Farm Management Platforms: Ecosystems like John Deere Operations Center that can benefit from Krishi Udaan’s crop health data and real-time updates.

1.6 How will the target users benefit from the solution

- **Real-Time Crop Health Monitoring**: Users receive up-to-date, data-driven insights on their crop health, helping them identify and address issues like pests, diseases, and nutrient deficiencies before they escalate.
- **Personalized, Accessible Advice**: Through the mobile app and AI chatbot, users get tailored guidance on irrigation, pest control, fertilization, and weather preparedness, ensuring they make informed decisions for each crop.
- **Drone-Powered Precision**: AI-enabled drones scan fields for early signs of stress, providing farmers with actionable data for timely interventions, minimizing waste, and improving crop yield.
- **Predictive Alerts for Better Planning**: Users get proactive alerts about potential risks (e.g., weather changes, pest outbreaks), allowing them to plan and safeguard their crops in advance.
- **Offline-Ready and Scalable Solution**: The platform works in low-connectivity areas with offline sync, ensuring farmers in remote regions can also benefit from advanced technologies without relying on stable internet access.

1.7 How will users discover and adopt the solution

- **Marketing Through Digital Platforms:** We'll spread the word through targeted ads on social media platforms like Facebook, Google, and YouTube, reaching farmers directly with engaging content, demo videos, and success stories. Farmers can quickly see the value in real-time crop monitoring.
- **Partnerships with Agri-Tech & Input Providers:** Collaborations with seed, fertilizer, and pesticide companies will ensure that Krishi Udaan's benefits reach farmers through trusted brands they already use. These partners can integrate our technology into their services, promoting it through their established channels.
- **Empowerment Through Agricultural Communities:** Our platform will be marketed via Farmer Producer Organizations (FPOs) and local agricultural community groups. These grassroots networks will help spread the word, demonstrating how Krishi Udaan boosts yields and safeguards crops.
- **Tie-ups with Government & Rural Development Agencies:** Partnerships with government initiatives like PM-Fasal Bima Yojana and Digital Agriculture Mission will amplify Krishi Udaan's reach, promoting the platform as an essential tool for modern farming in policydriven campaigns and rural outreach programs.
- **One-Stop Solution: Easy-to-Install App:** The Krishi Udaan mobile app will be simple to download and use. With a user-friendly interface in regional languages and offline access, farmers can easily adopt the solution without tech overwhelm—one app for all their crop health needs.

1.8 How the market segment will be affected by the solution

- Increased Crop Yields: Farmers will benefit from tailored recommendations to improve crop productivity, ensuring better harvests.
- Reduced Crop Loss: The early warning system will help prevent pest infestations, diseases, and adverse weather conditions, lowering crop loss.
- Cost Savings: By optimizing resource usage (water, fertilizers, pesticides), farmers will reduce costs and improve their ROI.
- Faster Adoption of Tech: With easy-to-use features and language options, the solution will encourage tech adoption among rural farmers, especially those who were previously hesitant about modern agricultural practices.

1.9 Opportunity Canvas

Opportunity Canvas- Krishi Udaan (3C53_1)

Users & Customers

- Farmers (Small, Marginal, and Large scale)
- Agribusiness companies needing data driven insights and government agencies aiming to promote sustainable farming prices

Business Challenges

- Overcoming traditional farnling skepticism
- Low technology literacy (particularly the elderlyor riderly or those in remote areas)

Solutions Today

- Low technology literacy (particularly among the elderly or those in rural areas)
- Manual monitoring (extensive, slow, subjective)
- High initial cost of tech adoption

Problems

- Inerficlent crop monitoring lead to delays in addressing crop issues
- Excessive use of water, fertilizers and pesticides increase costs Limited market access affecting pricing and sales
- High labor dependency impacts productivity

Solutions Ideas

- Government backed Agri-Tech awareness campaigns
- Partnerships with cooperatives & agribusinesses
- Government subsisdles for tech adoption

Adoption Strategy

- Government-backed Agri-Tech awareness campaigns
- Partnerships with cooperatives & agribusinesses

User Metrics

- Increase in Crop Yields.
- Reduction in Water & Fertilizer use.
- Adoption Rate among Farmers
- Government Policy and support

User Value

- Higher Yields: AI ensures optimal crop growth.
- Cost Efficiency: Reduces waste and unnecessary labor costs.
- Data-Driven Decisions: Farmers make smarter farming choices.

Budget

- Initial R&D: 5 Lakh for AI, drones, and IoT.
- Infrastructure: Cloud-based analytics, mobile platform, and blockchain integration.
- Marketing: Farmer outreach programs & government partnerships.

Business Metrics

- Break-Even Point: Recover 5 Lakhs infrastructure costs within 2 years.
- Revenue from Partnerships: Track income from B2B deals offering data insights, crop monitoring, analytics.
- Customer Acquisition Cost (CAC): Measure onboarding cost per farmer/client

2. CUSTOMER VALIDATION SURVEY

2.1 Form used for customer survey

Krishi Udaan: Smart Agricultural Drone Survey

Help us improve precision farming with AI-powered drones by sharing your insights!

Are you from farming background? *

- ◯ Yes
- ◯ No

Land Size (in acres):

- ◯ Less than 2
- ◯ 2 - 5
- ◯ 5 - 10
- ◯ More than 10

What is your average annual income from farming?

- ◯ Less than 5 Lakhs
- ◯ 5-10 Lakhs
- ◯ 10-15 Lakhs
- ◯ Above 15 Lakhs

According to you, how is crop health currently monitored? *

- ☐ Manual Inspection
- ☐ Satellite Imagery
- ☐ Drones
- ☐ AI-based Monitoring
- ☐ Other

What do you think are the major challenges faced in farming? *

- [ ] Crop Diseases & Pest Attacks
- [ ] Soil & Nutrient Deficiencies
- [ ] Weather Uncertainty
- [ ] High Cost of Inputs (Seeds, Fertilizers, Pesticides)
- [ ] Labor Shortages
- [ ] Low Market Prices

According to you, how often is technology (AI, apps, sensors, drones) used in farming? *

- ○ Never
- ○ Rarely
- ○ Sometimes
- ○ Frequently

Have you heard about AI-powered agricultural drones? *

- ○ YES
- ○ NO

Do you believe using drones for crop monitoring can reduce manual workload and increase efficiency?

- ○ Strongly Agree
- ○ Agree
- ○ Neutral
- ○ Disagree

What do you think about using a smart agricultural drone for farming? *

- ◯ YES
- ◯ NO
- ◯ MAYBE
- ◯ Depends on the functionality and features offered

What features would be most useful in a drone-based farming system? *

- ☐ Disease & Pest Detection
- ☐ Soil & Nutrient Analysis
- ☐ Real-time Weather Monitoring
- ☐ Automated Spraying & Irrigation Suggestions
- ☐ AI-powered Chatbot for Farming Guidance

Would you find an AI-powered chatbot that provides instant crop management advice, weather updates, and pest control suggestions useful? *

- ◯ Extremely useful
- ◯ Somewhat useful
- ◯ Not very useful
- ◯ Not useful at all

What do you think would be a reasonable cost for an AI-powered drone service per season? *

- ◯ Less than ₹5,000
- ◯ ₹5,000 - ₹10,000
- ◯ ₹10,000 - ₹20,000
- ◯ More than ₹20,000

If Krishi Udaan offers a subscription-based model with affordable pricing, would you consider subscribing? *

- ◯ Definitely
- ◯ Probably
- ◯ Not likely
- ◯ No, I prefer one-time purchase solutions

Do you have any suggestions for improving farming technology?

Long answer text

2.2 Size of the customer survey and documentary proof

The survey consists of responses from a total of 103 respondents.

The responses have been collected through the medium of a Google form that had been circulated in the university through WhatsApp and Social Media.

103 responses

View in Sheets

Summary Question Individual

Fig: QR code for survey form

Anadaan_be22@thapar.edu

vimmijain66@gmail.com

radhika.rajdev.rr@gmail.com

rrana1_be22@thapar.edu

sjain4_be22@thapar.edu

apriyadarshi_be22@thapar.edu

princejha2505@gmail.com

jainkriti2005@gmail.com

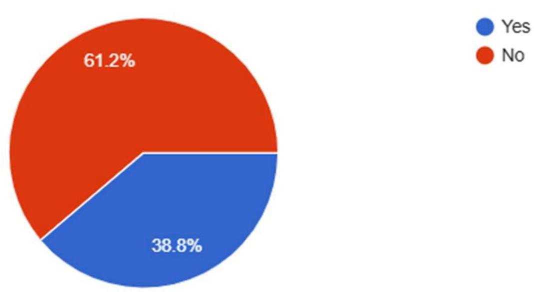
Khannaaryan1103@gmail.com

2.3 & 2.4 Result and Analysis of the customer survey response

Are you from farming background?

103 responses

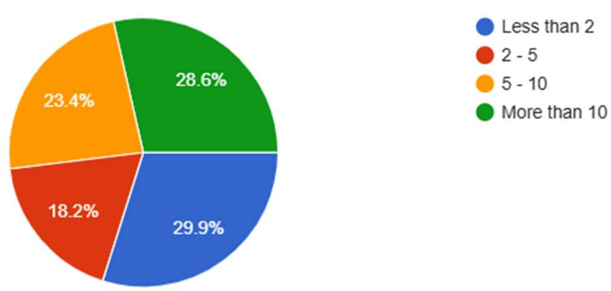
Copy chart



Land Size (in acres):

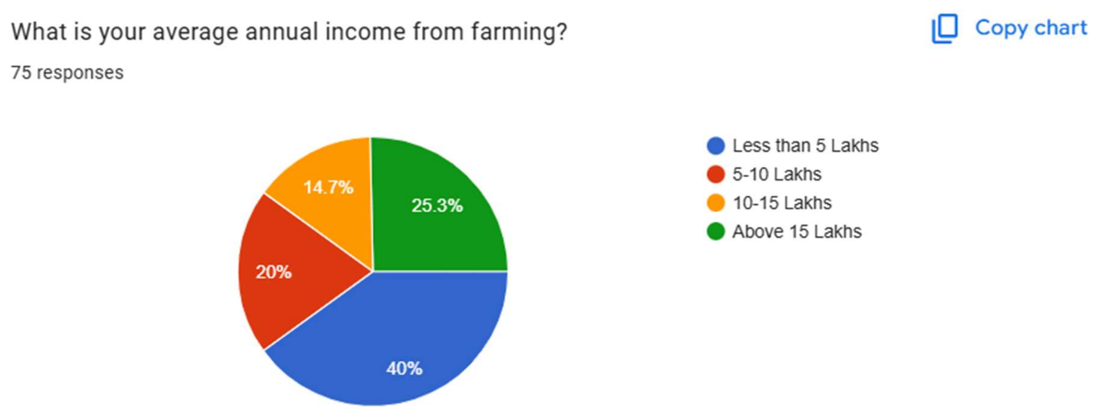
77 responses

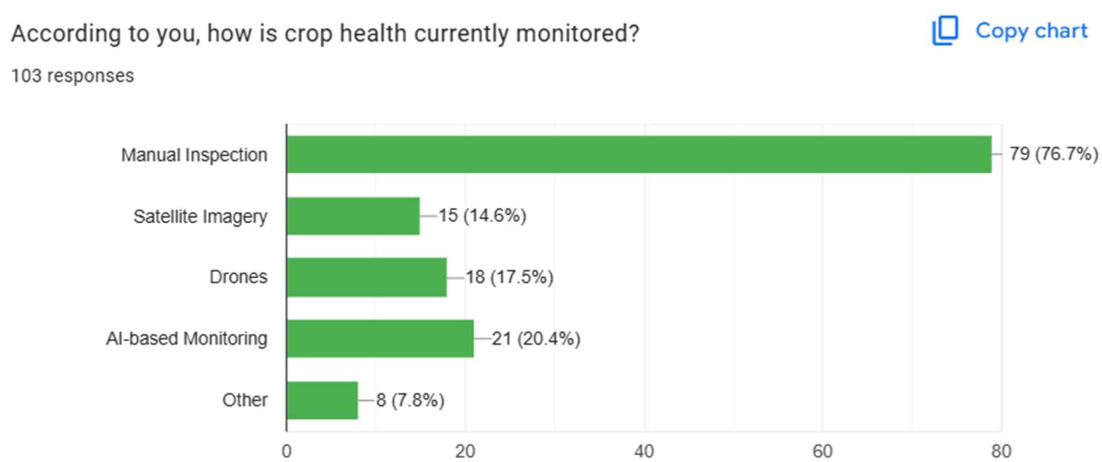
Copy chart



Analysis:

Our survey reached people from both farming and non-farming backgrounds, giving us a well-rounded understanding of how different groups perceive agricultural tech. Interestingly, around 61% of the respondents don't come from a farming background, which shows there's curiosity and interest even outside traditional farmer communities. Among those who do have a farming background, land sizes varied quite a bit with many owning less than 5 acres. This tells us that Krishi Udaan's focus on affordability and ease of use is right on track, especially for small and marginal farmers. It's clear that our solution needs to be simple, accessible, and informative to truly make an impact.

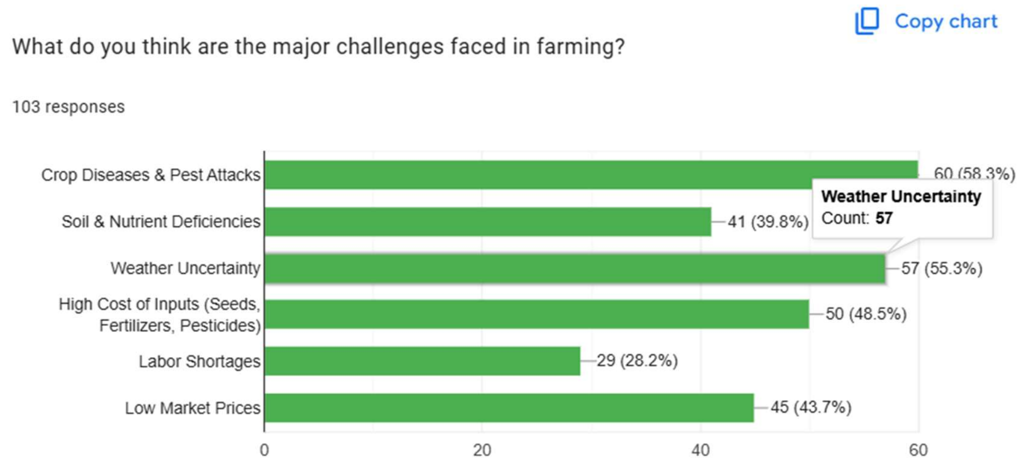




Analysis:

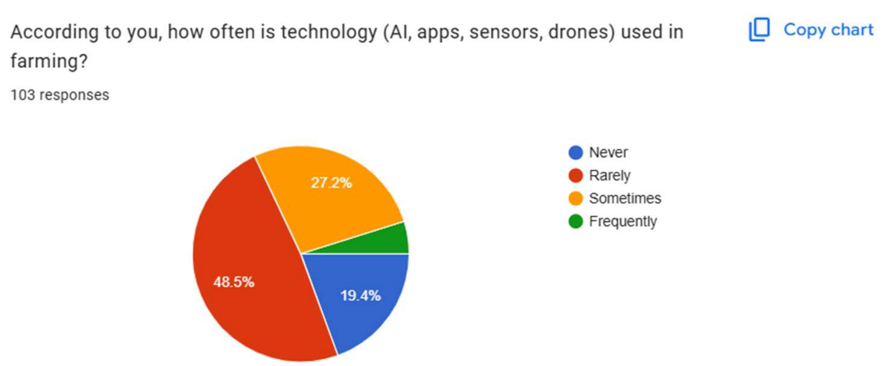
The responses show that a large number of farmers (40%) earn less than ₹5 lakhs annually from farming, reflecting the financial constraints many small and marginal farmers face. Only about 25% reported incomes above ₹15 lakhs, indicating that high-revenue farming is still limited to a smaller group. This reinforces the importance of making Krishi Udaan both affordable and impactful especially for those with lower incomes who need cost-effective solutions the most.

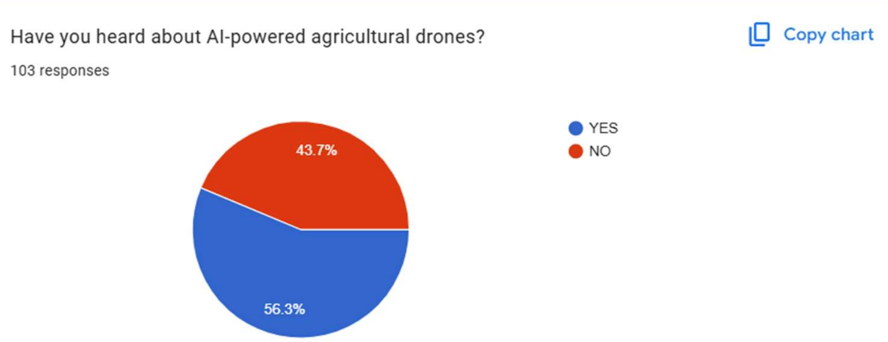
When it comes to monitoring crop health, the majority (about 77%) still rely on manual inspection. This highlights a major gap in tech adoption and presents a huge opportunity for innovation. While AI-based tools, drones, and satellite imagery are being used, they're still in the early stages of acceptance. This tells us that educating farmers and building trust in new technologies will be key to driving adoption of platforms like Krishi Udaan.



Analysis:

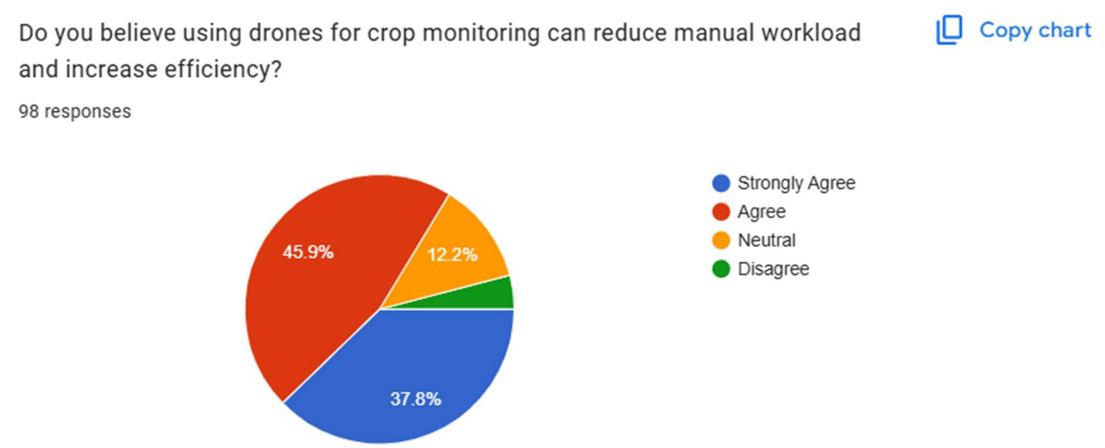
The survey reveals that the biggest challenges farmers face are crop diseases and pest attacks (58.3%) and weather uncertainty (55.3%). High input costs (48.5%) and low market prices (43.7%) also affect their income. Soil issues (39.8%) and labour shortages (28.2%) add to the burden, highlighting the need for affordable and tech-driven farming solutions.

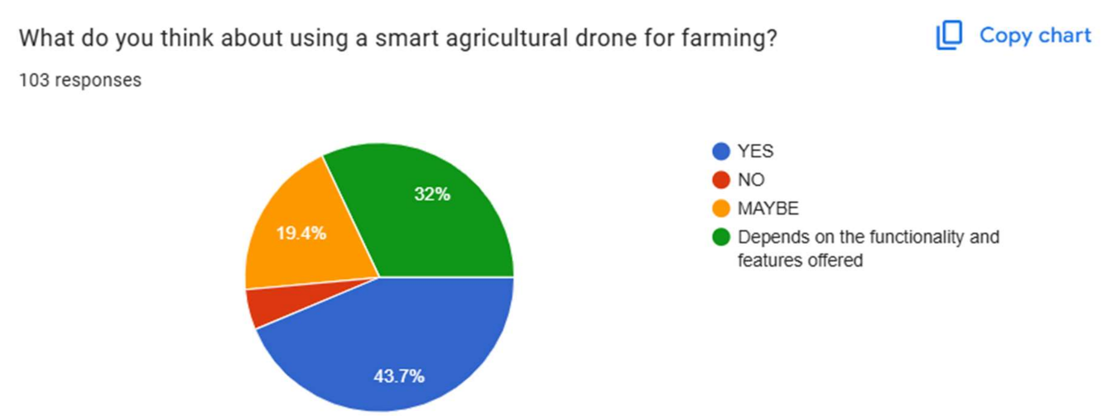




Analysis:

The data reveals that while a majority (48.5%) of respondents believe technology like AI, sensors, and drones is rarely used in farming, a promising 27.2% say it's used sometimes. Only a small fraction (4.9%) reported frequent use, showing that there's still a long way to go in terms of tech integration. However, it's encouraging to note that 56.3% of participants have heard about AI-powered agricultural drones, indicating growing awareness and potential openness toward adopting advanced farming solutions in the near future.

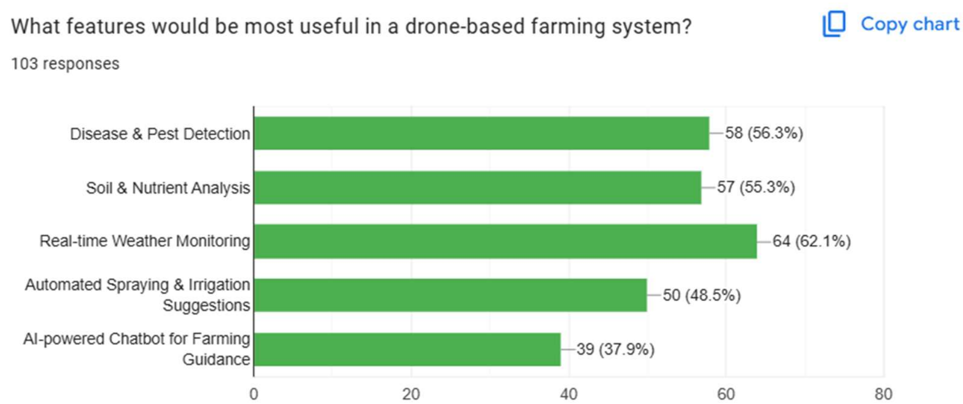




Analysis:

The survey indicates a strong belief in the effectiveness of drones for farming. A combined 83.7% of respondents (37.8% strongly agree and 45.9% agree) believe that drones can significantly reduce manual workload and boost efficiency in crop monitoring.

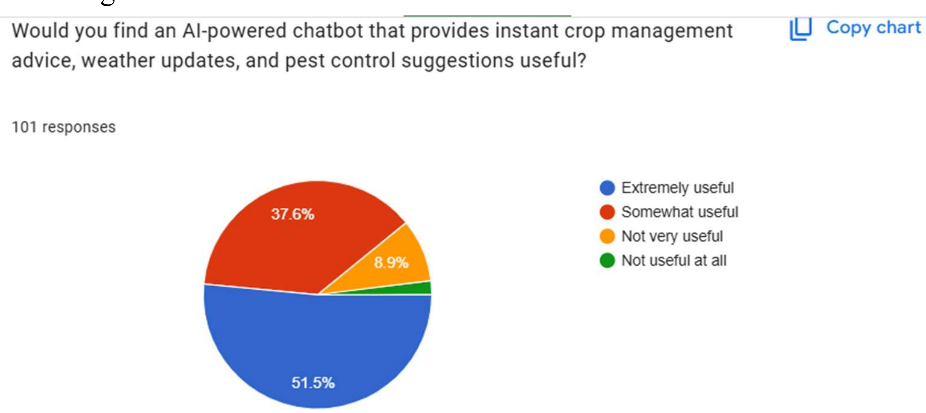
Furthermore, when asked about using smart agricultural drones, 43.7% responded positively, while 32% said their willingness depends on the features and functionality offered. Only a small percentage (4.9%) rejected the idea entirely. This reflects a growing openness toward drone-based solutions, provided they are practical and beneficial.

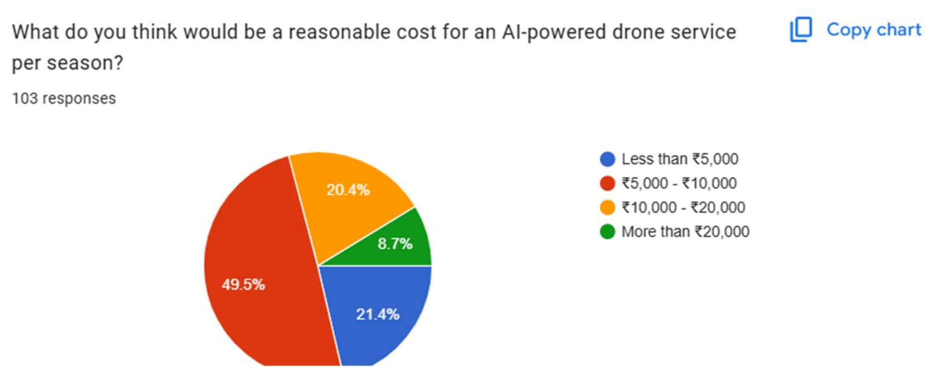


Analysis:

When asked about the most useful features in a drone-based farming system, respondents prioritized Real-time Weather Monitoring (62.1%) as the top choice. Close behind were Disease & Pest Detection (56.3%) and Soil & Nutrient Analysis (55.3%), highlighting a clear demand for precision and data-driven farming.

Other valued features included Automated Spraying & Irrigation Suggestions (48.5%) and AI-powered Chatbots for Farming Guidance (37.9%), showing that while automation and AI assistance are appreciated, farmers place greater importance on environmental and crop health monitoring.



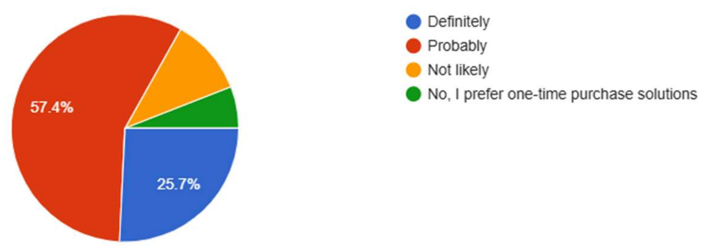


Analysis:

A majority of respondents (89.1%) found the AI-powered chatbot either extremely or somewhat useful, confirming strong interest in real-time, personalized farming support. In terms of pricing, nearly 71% of farmers preferred paying up to ₹10,000 per season for drone services, with ₹5,000–₹10,000 being the most acceptable range. This highlights a clear demand for practical, data-driven solutions at affordable price points.

If Krishi Udaan offers a subscription-based model with affordable pricing, would you consider subscribing? Copy chart

101 responses



Analysis:

When asked about subscribing to Krishi Udaan under an affordable subscription-based model, 83.1% of respondents said they would definitely (25.7%) or probably (57.4%) consider subscribing. Only a small segment preferred one-time purchases (5%) or were not likely to subscribe (11.9%). This indicates strong openness to a subscription model, provided the pricing aligns with perceived value and affordability.

Do you have any suggestions for improving farming technology?

30 responses

Farming in different types of geological terrain of incredible india will need Innovative farming technologies suitable for specific place and their culture. It can include:

1. Drip Irrigation for dry places like gujrat or east Maharashtra where soil quality is poor and issues like water scarcity is more prevalent will be an efficient water management technique that reduces water wastage and improves crop yields. Usage of AI drone in water management and determining soil health would help a lot in increasing overall yield.

2. In urban areas where people may have less space and opt for farming Vertical Farming can maximizes space by growing crops like leafy vegetables ,herbs in stacked layers sometimes under the controlled environment can bring upon much profit than usual. Using AI drones to determine and managing the ecosystem required for the vertical farming would bring considerable changes on existing system.

3 Aquaponics: in certain places like tropical areas or higher water levels AI Drones can monitor Combination of fish farming and plant cultivation, promoting sustainable practices and resources.

4. Live feeds and AI drones controlled from Mobile Apps for Farmers: Platforms providing weather updates, market prices, and agricultural advice.

5. Biotechnology: Development of drought-resistant and pest-resistant crop varieties.

2.5 Conclusion of the Survey

- **Strong interest in agri-tech:** The survey attracted a diverse audience, with over 60% from non-farming backgrounds, reflecting growing curiosity and acceptance of agricultural technology beyond traditional farming communities.
- **Target alignment with small farmers:** A significant portion of farming respondents owned less than 5 acres and earned under ₹5 lakhs annually, validating Krishi Udaan's focus on affordability and ease of use for small and marginal farmers.
- **Current reliance on manual methods:** About 77% of respondents still rely on manual crop monitoring, highlighting a major gap in tech adoption and a strong opportunity for AI and drone-based innovations.
- **Top farming challenges identified:** Farmers highlighted crop diseases (58.3%), weather uncertainty (55.3%), high input costs (48.5%), and low market prices (43.7%) as major pain points, reinforcing the need for tech-driven, cost-effective solutions.
- **Low current tech usage, but growing awareness:** While only 4.9% report frequent use of smart tech in farming, over 56% are aware of AI-powered drones, indicating rising awareness and readiness for adoption.
- **Positive perception of drones:** 83.7% believe drones can reduce workload and improve efficiency, and 75.7% are open to using them, depending on features and usability.
- **Preferred features:** Real-time Weather Monitoring (62.1%), Disease & Pest Detection (56.3%), and Soil & Nutrient Analysis (55.3%) were ranked highest, showing demand for precision and data-rich solutions.
- **High interest in chatbot support:** 89.1% of respondents found the AI chatbot helpful, validating its role in delivering personalized, real-time guidance.
- **Price sensitivity for drone services:** Nearly 71% prefer services priced below ₹10,000 per season, with ₹5,000–₹10,000 being the sweet spot—supporting the need for value-driven pricing strategies.
- **Openness to subscriptions:** 83.1% showed willingness to consider a subscription model, reinforcing the potential for a recurring revenue structure if value is consistently demonstrated.

3. Financial Model

3.1 Cost Structure

This cost structure represents the projected 1-year expenditure for Krishi Udaan, focused on initial operations in one regional cluster:

S.No	Expense Category	Description	Estimated Cost (₹)
1	Drone Procurement & Leasing	Acquisition and servicing of drone fleet	30,00,000
2	IoT Sensors & Hardware	Soil and weather sensors, installation, and calibration	10,00,000
3	Application Development	Mobile and web application, dashboard, backend services	15,00,000
4	AI & ML Model Training	Data labeling, model training, testing, and cloud computing	12,00,000
5	Salaries	Tech, field, sales, and support team	25,00,000
6	Advertisements & Promotions	Awareness drives, multilingual campaigns	8,00,000
7	Maintenance & Hosting	Server, app, drone servicing	5,00,000
8	Office & Utilities	Rent, electricity, internet	5,00,000
9	Miscellaneous	Travel, legal, contingency	5,00,000
	TOTAL		₹1,15,00,000

3.2 Revenue Structure

Krishi Udaan generates income through multiple channels:

- **Drone-as-a-Service (DaaS):** Charging ₹500/acre for aerial surveys and spraying. Assuming 2,000 acres/month → ₹1,00,000/month → ₹12,00,000/year
- **Subscription Plans for Farmers:** ₹500/month for real-time crop reports, chat support, and advisory tools. 500 farmers targeted → ₹30,00,000/year
- **B2B Sales (Agribusinesses / Gov Schemes):** Licensing data analytics dashboard/API → ₹30,00,000/year
- **Precision Spray Commissioning:** 10% margin on fertilizers/pesticides used via drones (partners). Estimated at ₹20,00,000/year
- **Total Revenue Estimate** (Year 1): ₹92,00,000

3.3 Profit & Loss Statement

Item	1st Year (₹)	2nd Year (₹)	3rd Year (₹)
Total Revenue	92,00,000	2,00,00,000	4,50,00,000
Drone & Hardware Ops	40,00,000	35,00,000	30,00,000
App & AI Dev	27,00,000	10,00,000	5,00,000
Salaries	25,00,000	35,00,000	45,00,000
Promotions	8,00,000	12,00,000	20,00,000
Hosting & Maintenance	5,00,000	7,00,000	10,00,000
Office & Utilities	5,00,000	6,00,000	7,00,000
Miscellaneous	5,00,000	5,00,000	5,00,000
Total Expenses	1,15,00,000	₹1,10,00,000	₹1,22,00,000
EBIT	-₹23,00,000	₹90,00,000	₹3,28,00,000
Net Profit	-₹23,00,000	₹90,00,000	₹3,28,00,000

3.4 Cash Flow Statement

Year	1st Year (₹)	2nd Year (₹)	3rd Year (₹)
Cash at Beginning	0	60,00,000	2,00,00,000
Cash from Ops	+97,00,000	+1,90,00,000	+3,28,00,000
Investment Raised	+60,00,000	0	0
Capex & Assets	-70,00,000	-50,00,000	-60,00,000
Returns to Inv.	-27,00,000	-1,00,00,000	-1,50,00,000
Net Cash Flow	+60,00,000	+1,40,00,000	+1,18,00,000
Cash at End	60,00,000	₹2,00,00,000	₹3,18,00,000

4. Reflection

Working on the **Krishi Udaan** project was a truly enriching experience that allowed us to explore the intersection of cutting-edge technology and agricultural sustainability. This project helped us understand the significant challenges faced by Indian farmers, especially in terms of crop health monitoring, and inspired us to think innovatively to develop real-time, AI-powered solutions.

We realized that for any startup or product, **understanding the problem from the user's perspective** is crucial. Visiting farms (even if virtually or through secondary data) and analysing current practices helped us recognize how outdated methods are affecting yields and sustainability. This grounded our project in real-world applicability and sharpened our problem-solving approach.

Teamwork played a central role once again. Each team member brought in unique ideas—be it in drone technology, AI implementation, or user interface design. Open communication and collaborative brainstorming helped us build a holistic system that integrates drones, analytics, and a chatbot to serve farmers better.

Through market analysis, we faced challenges like understanding **supplier dependencies for drone components**, identifying our **competitive edge** in a crowded market, and recognizing **buyer sensitivity to price and complexity**. This helped us appreciate the nuances of **Porter's Five Forces** in evaluating business feasibility and strategizing accordingly.

We also learned key entrepreneurial and professional skills such as:

- Developing a **user-centric solution** through empathy and data.
- The importance of **competitive differentiation** in a tech-heavy industry.
- **Pitching and presentation skills**, as we had to clearly communicate complex ideas to a non-technical audience.
- Applying **strategic thinking** through market forces analysis and value proposition building.

Some of the major hurdles we faced included:

- **High initial investment requirements** for drone development and testing.
- **Technical challenges** in integrating AI with drone imagery and building an intuitive chatbot.
- **Adoption barriers** among farmers due to affordability and digital literacy, which we addressed by emphasizing simplicity and regional language support.

In conclusion, **Krishi Udaan** not only helped us apply theoretical knowledge in a practical, impactful project but also instilled a deeper sense of purpose. Solving real-world agricultural problems with smart technology showed us how innovation can drive sustainability and social good.

Annexures

Opportunity Canvas:

Opportunity Canvas- Krishi Udaan (3C53_1)

Users & Customers

- Farmers (Small, Marginal, and Large scale)
- Agribusiness companies needing data driven insights and government agencies aiming to promote sustainable farming prices

Business Challenges

- Overcoming traditional farnling skepticism
- Low technology literacy (particularly the elderlyor riderly or those in remote areas)

Solutions Today

- Low technology literacy (particularly among the elderly or those in rural areas)
- Manual monitoring (extensive, slow, subjective)
- High initial cost of tech adoption

Problems

- Inerficlent crop monitoring lead to delays in addressing crop issues
- Excessive use of water, fertilizers and pesticides increase costs Limited market access affecting pricing and sales
- High labor dependency impacts productivity

Solutions Ideas

- Government backed Agri-Tech awareness campaigns
- Partnerships with cooperatives & agribusinesses
- Government subsisdles for tech adoption

Adoption Strategy

- Government-backed Agri-Tech awareness campaigns
- Partnerships with cooperatives & agribusinesses

User Metrics

- Increase in Crop Yields.
- Reduction in Water & Fertilizer use.
- Adoption Rate among Farmers
- Government Policy and support

User Value

- Higher Yields: AI ensures optimal crop growth.
- Cost Efficiency: Reduces waste and unnecessary labor costs.
- Data-Driven Decisions: Farmers make smarter farming choices.

Budget

- Initial R&D: 5 Lakh for AI, drones, and IoT.
- Infrastructure: Cloud-based analytics, mobile platform, and blockchain integration.
- Marketing: Farmer outreach programs & government partnerships.

Business Metrics

- Break-Even Point: Recover 5 Lakhs infrastructure costs within 2 years.
- Revenue from Partnerships: Track income from B2B deals offering data insights, crop monitoring, analytics.
- Customer Acquisition Cost (CAC): Measure onboarding cost per farmer/client

Business Model Canvas: